HUMPHREY VIRTUAL FARMS LLC // POST-AWARD TRL 8 & TRL 9 FUNDED ENGINEERING ROADMAP
Entity: Humphrey Virtual Farms LLC (‘CAGE: 1AHA8‘ | ‘SAM.gov UEI: S1M4ENLHTDH5‘)
Controlling Authority: Jeffery Humphrey, Founder, CEO, & Principal Sovereign Architect
Pre-Award Baseline Status: ‘TRL 7‘ Complete & Frozen (‘631 / 631‘ Modules, ‘83 / 83‘ Un
it Tests, Master Merkle Root: ‘cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e18e915
0949a‘)
Execution Condition: All Work Packages (‘WP-1‘ through ‘WP-6‘) execute **strictly post
-award** using settled funds from **Milestone #01 / Day-1 COTS License Release (‘18%‘ Set
up Hardware Bucket + ‘15%‘ TRL 8/9 Integration Bucket)**.

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1. Executive Boundary: Why Pre-Award Code Is Frozen at TRL 7

1. **Zero Uncompensated Pre-Award Custom Engineering:** The Project Ebony Sovereign C2 Opera
ting Console (‘26‘ Modules / ‘72‘ Verified Controls) is 100% complete at **Technology Re
adiness Level 7 (‘TRL 7‘)**. No further out-of-pocket pre-award code modifications are r
equired prior to contract award.
2. **Plug-In Modularity Without Core Rewrites:** All post-award customer adaptations (‘TRL
8‘ site hardware-in-the-loop binding and ‘TRL 9‘ multi-node base accreditation) attach as
 isolated plug-in drivers without modifying the sealed ‘631‘-module ‘c2_cockpit‘ core.

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2. Six Funded Post-Award Engineering Work Packages (Day 1 to Day 90)

| Work Package | Target TRL | Execution Window | Funding Source Bucket | Concrete Engineerin
g Deliverable (Executed Only After Award Funds Settle) |
| :--- | :--- | :--- | :--- | :--- |
| **‘WP-1: Bare-Metal Appliance Provisioning‘** | **TRL 8** | **Days 1 - 15** | ‘18.0%‘ Hard
ware Bucket (Milestone #01) | Procure customer-dedicated ruggedized edge server node(s) Tj T* ( ), dual NVMe h
001‘ - ‘0026‘ license keys. |
| **‘WP-2: 36.75ns SiC MOSFET Hardware Interlock Board‘** | **TRL 8** | **Days 10 - 30** | ‘
18.0%‘ Hardware Bucket (Milestone #01) | Assemble & bench-verify physical Silicon-Carbid
e (SiC) MOSFET solid-state DC bus breaker board (‘18.50ns‘ comparator + ‘12.00ns‘ gate
driver + ‘6.25ns‘ turn-off = ‘36.75ns‘ hardware guillotine) for customer robotic cells (
‘CLIN 0006‘). |
| **‘WP-3: Site-Specific PLC, SCADA, & CAN-Bus Binding‘** | **TRL 8** | **Days 15 - 45** | ‘
15.0%‘ TRL 8/9 Integration Bucket | Map customer’s site-specific Modbus-TCP / DNP3 substat
ion relays (‘CLIN 0003‘), 480V grain silo contactors (‘CLIN 0009‘), or SAE J1939 tacti
cal vehicle CAN-Bus interfaces (‘CLIN 0008‘) into isolated plug-in adapter profiles. |
| **‘WP-4: UAS / Drone Optical Feed & Title 61 GIS Hookup‘** | **TRL 8** | **Days 20 - 45**
| ‘15.0%‘ TRL 8/9 Integration Bucket | Connect customer’s local UAS ground control station
 RTSP/optical feeds (‘CLIN 0007‘) and county/base GIS shapefiles (‘CLIN 0010‘) into th
e air-gapped GLI and Title 61 compliance compiler. |

| **‘WP-5: Customer UI Polish & Role-Based Watch-Floor Views‘** | **TRL 8/9** | **Days 30 - 60** | ‘15.0%‘ TRL 8/9 Integration Bucket | Tailor customer-requested dashboard logos, unit callsigns, watch-floor operator permissions, and automated shift-brief PDF templates (‘CLIN 0022‘) to match the awarding command’s SOP. |
| **‘WP-6: DoD RMF / eMASS ATO & Multi-Node Fleet Rollout‘** | **TRL 9** | **Days 45 - 90** | ‘15.0%‘ TRL 8/9 Integration Bucket | Generate site-specific DoD Risk Management Framework (RMF / NIST 800-53 Rev 5) STIG checklists, complete local Authorizing Official (AO) accreditation, and activate 50-seat Field Academy cohorts (‘CLIN 0011‘). |

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3. Architectural Guarantee: Zero-Rewrite Plug-In Adapter Pattern

Every post-award Work Package (‘WP-1‘ through ‘WP-6‘) is architected to load via ‘HVF_SITE_ADAPTER_MANIFEST.json‘ and isolated hardware driver classes, guaranteeing:
* **0 Bytes Modified** inside the sealed ‘631‘ ‘c2_cockpit‘ core modules (‘cc1ee1c8...949a‘).
* **100% Backwards Compatibility** with all ‘83 / 83‘ unit test suites and ‘1,420 / 1,420‘ Field Academy questions.
* **100% Customer-Funded Execution:** No hardware ordering or site travel occurs until the Day-1 invoice clears into Humphrey Virtual Farms LLC’s bank account.